

VPOS FTC Field Support, Installation & Commissioning Manual

Audience: Gilbarco/authorized field technicians, commissioning engineers, and production support

Purpose: Manual for installing, configuring, commissioning, supporting, and handing over VPOS FTC on a DOMS controller.

Production model: Secure package installed on DOMS; no SSH, terminal, or field command execution is required.

This is the primary on-site support manual. It is intentionally self-contained so a technician can complete installation, configuration, commissioning, first-line diagnosis, and handover without relying on R&D Team.

Field operating rule

Production adjustments are made through the VPOS application and approved DOMS package-management workflow. Do not attempt to execute application scripts, or use SSH as part of normal installation or support.

1. Production deployment model

VPOS FTC is delivered as an approved Node.js 22 package for the target DOMS controller.

DOMS controller	Production package	Architecture
CPB-579	cpb-579-node22.pkg	ARM64
CPB-539	cpb-539-node-22.pkg	ARMv7I

The package contains the application runtime and the required production defaults. It is installed on the DOMS using the approved DOMS package installation mechanism and runs as a managed, secured application.

The field support model therefore assumes:

- configuration changes are made from supported VPOS administration screens
- package start, stop, upgrade, and rollback use the approved DOMS package-management controls

If a production issue appears to require shell access, manual database manipulation, or application-file modification, stop and escalate to R&D. That is outside the supported field procedure.

2. Configuration model

VPOS is designed so station-specific operational configuration can be maintained from the application rather than from environment files.

Field-adjustable configuration includes the applicable station settings for:

- site identity, country, currency, and timezone

- products and grades
- tanks and tank-grade relationships
- pumps, nozzles, and nozzle-to-tank mappings
- DOMS/JPL connection settings
- forecourt operating settings
- approved forecourt pricing
- printer configuration
- ATG/tank polling settings
- proxy/fiscal configuration
- Country specific fiscal settings where applicable
- other administrator-owned station configuration exposed by the application

Production packages contain safe runtime defaults for values that do not require site-specific adjustment. The application persists supported site-specific configuration in its station data model.

Some security material, such as certificates or secure credentials, may be provisioned by the approved package/deployment process. Field staff should manage only the application settings and references exposed through authorized VPOS screens.

3. Roles and access required on site

The technician should have an authorized `administrator` account for commissioning and technical support.

Administrator functions include the additional controls required for installation validation and diagnosis, such as:

- Setup Wizard
- Forecourt Monitor
- Device Status
- Diagnostics
- Print Jobs
- Proxy Settings
- Tanzania Fiscal where applicable
- Runtime Control
- Maintenance
- Station Settings / Station Config
- Printers
- Products and Product Categories
- Users where approved

Do not use or share another person's administrator credentials. Managers should continue to use manager accounts for daily operations.

4. Information required before installation

The deployment/change record should contain the following before work begins.

Package and site

- station/site name and identifier
- DOMS controller model: CPB-579 or CPB-539
- currently installed VPOS version/package where replacing an existing version
- approved target package and version
- approved rollback package/version
- maintenance window/change reference
- country, currency, and timezone
- local site contact and escalation contact

DOMS/PSS/JPL

- production DOMS/PSS host or address as presented to VPOS
- approved JPL TCP port; default is normally 8888
- site-assigned JPL POS ID
- approved operation mode: supervised or unsupervised
- approved access-code/capability expectations
- country code and expected JPL/PSS version where required
- TLS requirements where enabled

Forecourt

- pump identifiers
- nozzle identifiers
- tank identifiers
- products/grades
- nozzle-to-tank relationships
- approved forecourt prices

Peripherals and fiscal services

- approved printer endpoint/details
- approved proxy configuration
- country-specific fiscal prerequisites
- Tanzania fiscal/device information where applicable

Do not discover production configuration by trial and error when the information should already exist in the approved PSS configuration.

5. Pre-installation checks

Before changing the station:

1. Confirm the DOMS controller model.
2. Confirm the package matches that controller exactly.
3. Confirm the controller's date, time, and timezone are correct.
4. Confirm the current station is in a safe state for the maintenance window.
5. Record the currently installed VPOS package/version if applicable.
6. Record known pre-existing faults so they are not attributed to the new release later.
7. Confirm the rollback package is available through the approved DOMS package process.
8. Confirm the station has the approved network path to DOMS/PSS, printer, and proxy/fiscal services.
9. Confirm the site configuration and physical forecourt information are available.
10. Confirm PostgreSQL is enabled. (DOMS FCC Web App step 5.10 Database Maintenance)

6. Install or upgrade the package on DOMS

Use the standard DOMS package install process. (5.2 Software Update using FCC Web App)

CPB-579

Install the latest `cpb-579-node22.pkg` package.

CPB-539

Install the latest `cpb-539-node-22.pkg` package.

Installation sequence

1. Notify the site manager maintenance will commence and pumps should be turned offline.
2. Record the old package/version if upgrading.
3. Ensure firmware is up to date (Node 22).
4. Select the correct package.
5. Install or upgrade the package using the DOMS package installer.
6. If VPOS is not started automatically, Start the VPOS package through VPOS Admin.
7. Open VPOS from an authorized browser/client and confirm that the application is reachable on port `3080`.
8. Continue with startup validation and initial configuration.
9. Sign in with the authorized administrator account.
10. Continue with in-app configuration.

Do not copy individual runtime files, modify package contents, or attempt to repair an installation by changing files on the DOMS.

7. First startup expectations

The package performs its own startup/bootstrap work. This includes the application database bootstrap, ordered schema migrations, HTTP service startup, runtime initialization, forecourt connection initialization, and worker startup.

A technician does not manually run database migrations.

During first startup or upgrade:

- Allow the application to complete startup
- Use the VPOS UI and administrator diagnostics to determine whether startup is healthy `/api/livez`
- if the application reports a migration/bootstrap failure, capture the visible error and escalation details rather than attempting SQL repair
- if an upgrade cannot reach a stable application state inside the change window, follow the approved package rollback procedure

8. Initial application checks

After the package starts:

1. Confirm the login page loads normally.
2. Sign in as administrator.
3. Confirm the Dashboard renders without a global startup failure.
4. Open **Diagnostics** and note any critical application/runtime failure.
5. Open **Device Status** and identify any configured dependency that is unavailable.
6. Open **Forecourt Monitor** and determine whether the DOMS/JPL session is disconnected, connecting, stable, or repeatedly reconnecting.
7. Open **Print Jobs** only if printer testing or troubleshooting is required.
8. Confirm the application can navigate between administration pages without repeated server errors.

Do not treat a single loaded page as proof that forecourt, printing, database, proxy, and fiscal functions are all healthy.

Live

Refresh logs

Download selected

Clear selected from DB

Open via API

SEL	FILENAME	SIZE	MODIFIED
☐	application.log	195.31 KB	2026-09-18T08:05:21.839Z
☐	doms-jpl.log	195.31 KB	2026-09-18T08:04:43.132Z
☐	application-errors.log	195.31 KB	2026-09-18T07:49:40.455Z

Preview: application-errors.log

[2026-09-16T01:41:26.244Z] [6133] [ERROR] [ERROR] [[tanzaniaDailyTotalsWorker] tick failed] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","error":{"name":"Error",

[2026-09-16T02:23:25.846Z] [6133] [ERROR] [ERROR] [[tanzaniaDailyTotalsWorker] tick failed] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","error":{"name":"Error",

[2026-09-16T03:50:26.194Z] [6133] [ERROR] [ERROR] [[tanzaniaDailyTotalsWorker] tick failed] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","error":{"name":"Error",

[2026-09-16T04:32:26.089Z] [6133] [ERROR] [ERROR] [[tanzaniaDailyTotalsWorker] tick failed] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","error":{"name":"Error",

[2026-09-16T05:34:12.003Z] [6133] [ERROR] [ERROR] [[jplTcp] client disconnect] {"args":[]}

[2026-09-16T05:34:12.007Z] [6133] [ERROR] [ERROR] [[jplTcp] connection lost] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","reason":"disconnect","reconnectAttempts

[2026-09-16T05:34:12.063Z] [6133] [WARN] [WARN] [[jplTcp] scheduling reconnect] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","delayMs":1000,"reconnectAttempts":1,

[2026-09-16T05:34:13.236Z] [6133] [ERROR] [ERROR] [[JPL-ADAPTER] connect/logon failed] {"host":"192.168.68.123","port":8888,"stationId":"3ec21997-0c3b-44bf-a492-d23607c

[2026-09-16T05:34:13.254Z] [6133] [ERROR] [ERROR] [[jplTcp] connection lost] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","reason":"connect/logon_failed","error":

[2026-09-16T05:34:13.490Z] [6133] [WARN] [WARN] [[jplTcp] scheduling reconnect] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","delayMs":2000,"reconnectAttempts":2,

[2026-09-16T05:34:15.522Z] [6133] [ERROR] [ERROR] [[JPL-ADAPTER] connect/logon failed] {"host":"192.168.68.123","port":8888,"stationId":"3ec21997-0c3b-44bf-a492-d23607c

[2026-09-16T05:34:15.528Z] [6133] [ERROR] [ERROR] [[jplTcp] connection lost] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","reason":"connect/logon_failed","error":

[2026-09-16T05:34:15.563Z] [6133] [WARN] [WARN] [[jplTcp] scheduling reconnect] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","delayMs":4000,"reconnectAttempts":3,

[2026-09-16T06:32:07.661Z] [6287] [WARN] [diag] received SIGTERM (pid=6287 rss_mb=143 heap_used_mb=54)

[2026-09-16T07:06:07.157Z] [26218] [WARN] [diag] received SIGTERM (pid=26218 rss_mb=129 heap_used_mb=53)

[2026-09-18T06:41:42.414Z] [12199] [ERROR] [ERROR] [[tanzaniaDailyTotalsWorker] tick failed] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","error":{"name":"Error",

[2026-09-18T06:41:46.053Z] [12199] [ERROR] x AuthError: Unauthorized

[2026-09-18T06:42:38.314Z] [12199] [ERROR] [ERROR] [[tanzaniaDailyTotalsWorker] tick failed] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","error":{"name":"Error",

[2026-09-18T06:43:40.356Z] [12199] [ERROR] [ERROR] [[tanzaniaDailyTotalsWorker] tick failed] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","error":{"name":"Error",

[2026-09-18T06:45:38.542Z] [12199] [ERROR] [ERROR] [[tanzaniaDailyTotalsWorker] tick failed] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","error":{"name":"Error",

[2026-09-18T06:47:40.580Z] [12199] [ERROR] [ERROR] [[tanzaniaDailyTotalsWorker] tick failed] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","error":{"name":"Error",

[2026-09-18T06:52:38.405Z] [12199] [ERROR] [ERROR] [[tanzaniaDailyTotalsWorker] tick failed] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","error":{"name":"Error",

[2026-09-18T07:00:40.255Z] [12199] [ERROR] [ERROR] [[tanzaniaDailyTotalsWorker] tick failed] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","error":{"name":"Error",

[2026-09-18T07:17:38.561Z] [12199] [ERROR] [ERROR] [[tanzaniaDailyTotalsWorker] tick failed] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","error":{"name":"Error",

[2026-09-18T07:49:40.368Z] [12199] [ERROR] [ERROR] [[tanzaniaDailyTotalsWorker] tick failed] {"stationId":"3ec21997-0c3b-44bf-a492-d23607c4394b","error":{"name":"Error",

9. Setup Wizard

For a new station, approved recommissioning, or configuration recovery, open:

Setup & Configuration → Setup Wizard

Use the wizard to complete and verify the station configuration.

For an upgrade of a working commissioned station, do not overwrite existing values simply because the wizard is available. Confirm that the previously approved configuration has been retained.

10. Site profile

Configure and verify:

- site name
- tax number where applicable
- address
- country
- currency
- timezone

Country and timezone are operational settings. Incorrect values can affect fiscal behavior, reporting, daily totals, and transaction date interpretation.

11. DOMS/JPL forecourt connection

In the Setup Wizard/forecourt connection section, configure the production DOMS/JPL values provided by the site commissioning record.

Typical fields include:

Setting	Guidance
JPL host	Use the approved production DOMS/PSS host (127.0.0.1).
JPL port	Use the approved site port; the normal default is 8888 .
POS ID	Use the unique site-assigned POS ID.
Operation mode	Must match the approved site model.
Access code/capabilities	Must match the approved integration scope.
Country code	Use the approved site/vendor value.
POS/JPL version expectations	Keep aligned with the approved DOMS/PSS version.
Heartbeat/dead-connection settings	Change only where the approved integration requires it.
Unsolicited subscriptions	Keep the required station subscriptions enabled.
TLS settings	Use the approved security configuration where TLS is required.

Use the in-app **Test JPL settings** function before enabling production operations.

Do not use price changes, maintenance actions, or dispenser controls merely to test connectivity.

0b) Forecourt connection (required)

This setup path is now JPL-only. Production deployments do not rely on filesystem environment variables; the values saved here override the server defaults through `station_kv`.

Runtime target

jpl_tcp is the active and supported forecourt runtime.

JPL TCP host

JPL TCP port

JPL POS ID

Operation mode

JPL access code

JPL country code

JPL unsolicited DR seconds

JPL heartbeat interval (ms)

Use 15000 ms or lower. DOMS recommends sending traffic at least every 15 seconds.

JPL dead connection timeout (ms)

JPL expected minimum version

Use 30000 ms for field acceptance. It must be greater than the heartbeat interval, so 15000 ms is blocked when heartbeat is also 15000 ms.

Unsolicited subscription flags

MFDR subscription flags

Status update code

Bootstrap snapshot

Warn depth (supervised)

Critical depth (supervised)

Warn age min (supervised)

Critical age min (supervised)

Warn depth (unsupervised)

Critical depth (unsupervised)

Warn age min (unsupervised)

Critical age min (unsupervised)

JPL settings verified

Connected to 192.168.68.123:8888 and accepted access code

***,RI,UNSO_INSTSTA_1,UNSO_TRBUFSTA_3,UNSO_TGSTA_1,UNSO_DELIVSTA_1,UNSO_PRISTA_1,UNSO_FPSTA_3:MFDR=05. Status updates: OK; optional all-pump snapshot: OK.

[Test JPL settings](#)[Save forecourt settings](#)

12. Reconcile VPOS with PSS configuration

Use the application's PSS configuration verification/reconciliation capability where available.

Confirm that VPOS and the physical/PSS configuration agree on:

- pumps
- nozzles
- tanks
- products/grades
- mappings
- applicable forecourt settings

If the PSS itself is wrong, correct it through the approved PSS Configurator process and then rerun VPOS verification. Do not create a deliberately incorrect VPOS mapping to compensate for a PSS configuration defect.

DOMS configuration reconciliation

OK

Export JSON

Refresh reconciliation

DOMS configuration reconciliation

Compare FTC mappings with the latest DOMS/JPL snapshots, then apply or roll back FTC-side mapping changes only.

CONFIGURED PUMPS

4

CONFIGURED NOZZLES

12

CONFIGURED TANKS

4

ISSUES

0

OBSERVED FPIDS

4

OBSERVED TGIDS

0

OBSERVED TANKIDS

0

SUGGESTIONS

0

BLOCKING ISSUES

0

INSTALL STATUS

9/18/2026, 11:04:15 AM

MAPPING CHANGES

0

Mapping issues

0 errors 0 warnings

No reconciliation issues found from the currently available snapshots.

Latest DOMS install groups

10H

1, 2, 3, 4

4

50H

No device IDs

0

40H

1, 2, 3, 4

4

37H

No device IDs

0

2AH

No device IDs

0

FTC-side remediation suggestions

0 suggestions

Bulk FTC-side mapping review/apply

not reviewed

FTC mapping change history

0 changes

13. Pumps and nozzle-to-tank mapping

Use Setup Wizard, **Pumps**, **Pump Settings**, and **Forecourt Setup** as appropriate.

For every active nozzle:

1. identify the physical pump/nozzle
2. match it to the DOMS/PSS identifier
3. select the physical tank supplying that nozzle
4. verify the corresponding product/grade
5. save only after the mapping is confirmed
6. recheck the forecourt view after saving

Do not use trial-and-error mapping on a live station.

14. Products, tanks, and grades

Use the supported configuration pages to confirm:

- each active fuel grade has the correct product record
- each physical tank exists and is active where appropriate
- each tank is assigned the correct grade/product
- tank identifiers match the approved site/PSS topology

- ATG polling settings are correct where wet-stock integration is used

Do not apply arbitrary stock adjustments to hide a mapping or ATG issue.

15. Printer configuration

Use the Setup Wizard or **Setup & Configuration → Printers**.

Configure the approved printer and perform the in-app tests available for:

- printer connectivity
- transaction receipt output
- report output

Confirm physical output, paper/layout, station identity, and required fiscal content.

For troubleshooting, administrators can use **Print Jobs** to see whether a print request is queued, processing, failed, or completed.

Print Jobs

Monitor queued printer work, investigate failures, retry failed jobs, and clear queue history.

Clear all

Refresh

PENDING
0

PROCESSING
0

DONE
0

FAILED
22

Queue history

All statuses

All job types

CREATED	TYPE	PRINTER	SOURCE	STATUS	ATTEMPTS	COMPLETED	ERROR	ACTIONS
9/10/2026, 4:35:11 PM	setup.check_printer_page_width 1f50b7fb...	default	—	FAILED	1/3	9/10/2026, 4:35:11 PM	connect ECONNREFUSED 192.168.68.100:9100	Retry Clear
9/10/2026, 4:32:54 PM	print.receipt 38513c21...	default	Transaction 6185c091...	FAILED	3/3	9/10/2026, 4:33:02 PM	No printer configured (set device_configs...	Retry Clear
9/10/2026, 4:32:53 PM	print.receipt 57530df5...	default	Transaction aa2f4bdd...	FAILED	3/3	9/10/2026, 4:33:02 PM	No printer configured (set device_configs...	Retry Clear
9/10/2026, 4:32:53 PM	print.receipt d4b9dbe1...	default	Transaction 2c567104...	FAILED	3/3	9/10/2026, 4:33:02 PM	No printer configured (set device_configs...	Retry Clear
9/10/2026, 4:32:52 PM	print.receipt 57b7a3d3...	default	Transaction 39f50ed1...	FAILED	3/3	9/10/2026, 4:33:02 PM	No printer configured (set device_configs...	Retry Clear
9/10/2026, 4:32:52 PM	print.receipt 049204d4...	default	Transaction 5a550058...	FAILED	3/3	9/10/2026, 4:33:02 PM	No printer configured (set device_configs...	Retry Clear

16. Forecourt Monitor and connection acceptance

Open **Forecourt Monitor** and confirm:

- the JPL session establishes successfully
- the session remains stable rather than reconnecting repeatedly
- expected pump state is visible

- transaction-buffer/state information is current
- tank/wet-stock information is visible where configured
- no persistent protocol/authentication rejection is shown

If the connection is unstable, correct the approved settings through the application. Do not attempt network or process troubleshooting through the DOMS shell because no shell access is part of the production support model.

JPL diagnostics

unknown

JPL diagnostics

Protocol health, heartbeat freshness, correlation mode, and recent rejects.

unknown

Refresh diagnostics

CONNECTION

online

CONNECTED

yes

LAST MESSAGE AGE

14s

LAST HEARTBEAT AGE

8s

JPL VERSION

470-02-1.10

SECURE TRANSPORT

no

CORRELATION

unknown

REQUEST MODE

—

No protocol health issues currently reported.

Controller flags

{}

Subscriptions and dispatch

{
 "lastDisconnectReason": null,
 "nextReconnectAt": "—"
 }

Recent protocol rejects

20

RejectMessage_resp_00H

9/16/2026, 1:14:53 AM

RejectMessage_resp_00H

{
 "schemaVersion": 1,
 "sourceFieldCount": 7
 }

RejectMessage_resp_00H

9/16/2026, 12:00:01 AM

RejectMessage_resp_00H

{
 "schemaVersion": 1,
 "sourceFieldCount": 7
 }

RejectMessage_resp_01H

9/15/2026, 3:15:52 PM

RejectMessage_resp_01H

{
 "schemaVersion": 1,
 "sourceFieldCount": 7
 }

Recent JPL events

FpStatus_resp_00H

9/18/2026, 11:25:21 AM

FpStatus_resp_00H

FpStatus_resp_00H

9/18/2026, 11:25:21 AM

FpStatus_resp_00H

FpStatus_resp_00H

9/18/2026, 11:25:21 AM

FpStatus_resp_00H

FpStatus_resp_00H

9/18/2026, 11:25:21 AM

FpStatus_resp_00H

clear_FcServiceMsg_resp_00H

9/18/2026, 11:24:38 AM

clear_FcServiceMsg_resp_00H

FcServiceMsg_resp_00H

9/18/2026, 11:24:37 AM

FcServiceMsg_resp_00H

17. Forecourt pricing

Open **Forecourt Pricing** and confirm that displayed grades/products and prices match the approved site price set.

A price change is a controlled operational action. Use the site’s approved authorization and verification process.

Do not change a price as a connectivity test.

18. Proxy and fiscal services

Open **Fiscal Services** → **Proxy Settings** and verify the approved configuration/status.

Determine whether fiscal failures are:

- local transaction/receipt generation
- station-to-proxy connectivity/configuration

- proxy/cloud processing
- country fiscal service/device processing

Do not repeatedly resubmit the same transaction without understanding its current state.

19. Tanzania stations

For Tanzania, complete the administrator setup under **Tanzania Fiscal** and the applicable daily-total workflow.

Important operating semantics:

- the invoice number/daily counter can be scoped to the originating transaction date
- the Z number/invoice date can be scoped to the first fiscalization or pre-fiscalization assignment date
- a delayed transaction can therefore legitimately contain two different business/fiscal dates
- opening a receipt preview may persist the receipt identity where pre-fiscalization printing is enabled
- retries should reuse the persisted identity rather than create a new one

Do not manually alter invoice numbers, daily counters, Z numbers, fiscal identifiers, or database state to resolve a single transaction discrepancy.

20. Final setup verification

Before finalizing setup, confirm:

- site profile correct
- correct country/currency/timezone
- JPL settings tested successfully
- Forecourt Monitor stable
- PSS/VPOS reconciliation accepted
- products/grades correct
- tanks correct
- every active nozzle mapped correctly
- approved prices correct
- printer configured and tested
- proxy/fiscal status accepted where required
- Tanzania configuration complete where applicable

Only finalize after the configuration represents the physical site.

21. Production acceptance test

Perform the site-approved controlled acceptance workflow.

Minimum acceptance evidence:

- application login and Dashboard healthy
- Diagnostics has no unresolved critical startup failure
- Forecourt Monitor shows a stable DOMS/JPL session
- expected pumps/nozzles/tanks are visible
- pump status changes are reflected correctly
- transaction ingestion is visible in VPOS
- a known transaction can be located
- receipt preview/generation works
- physical printing works
- fiscal/proxy flow works where applicable
- tank level/wet-stock information is plausible where configured
- approved prices are correct
- no unexplained repeated reconnect, queue, or runtime failure is present

If a controlled fuel transaction is required, follow the station test-sale procedure and reconcile it afterward.

22. Controlled restart and persistence check

If the release procedure requires a restart, use only the approved DOMS application/package control or the authorized in-app runtime control provided for that purpose.

After restart verify:

- VPOS becomes reachable again
- administrator sign-in works
- saved station configuration remains present
- the JPL session reconnects and stabilizes
- pumps/tanks return to current state
- printer remains usable
- proxy/fiscal status returns to normal
- no new startup/migration error is visible in Diagnostics

Do not accept a deployment that works only until the first controlled restart.

23. Troubleshooting model

Field diagnosis should begin in the application. Preserve evidence before restarting or changing configuration.

Application cannot be opened

Check the DOMS package-management interface first:

- correct package installed for the controller
- package is enabled/running according to DOMS

- installation did not report an error
- no rollback or failed-upgrade state is shown

If the package cannot start and the approved UI does not provide enough evidence, escalate. Do not seek shell access as the next field step.

Application opens but station is degraded

Use **Diagnostics** and **Device Status** to identify the affected component. Determine whether the issue is database/startup, forecourt, printer, proxy/fiscal, or another configured dependency.

Many/all pumps stale

Use **Forecourt Monitor** and **Diagnostics**. Check the configured JPL host/port/POS ID/operation mode/TLS values through the application. Look for reconnect churn or an authentication/protocol rejection.

Do not remap every pump when the entire forecourt connection is failing.

One pump/nozzle wrong

Compare the physical dispenser, PSS identifier, VPOS pump/nozzle, tank mapping, and product/grade. Correct only the confirmed configuration error through the appropriate VPOS or PSS configuration tool.

Fiscalization failures

Use the transaction state plus **Proxy Settings** and country fiscal screens. Determine whether one transaction or many transactions are affected. Preserve the transaction ID and visible error/reference.

Receipt exists but does not print

Use **Print Jobs** and printer status. Establish whether the first print is queued, failed, or complete before retrying.

Tank data stale or implausible

Check whether the forecourt/JPL connection is healthy first, then confirm tank identifiers, mappings, ATG polling state, and current physical/site evidence.

Reports or dates appear wrong

Check the station timezone and report filters in the application. Do not change timezone merely to make a report total align.

24. Support evidence and escalation

Capture the following before escalation:

- station identifier

- DOMS controller model
- installed VPOS package/version
- local time and timezone
- first failure time and last known good time
- exact application page/workflow
- affected pump/nozzle/tank/transaction/receipt/print-job identifier
- visible error text and request/reference ID where shown
- whether one item or the whole station is affected
- screenshots of relevant Diagnostics, Forecourt Monitor, Device Status, Print Jobs, or transaction state
- recent package upgrade, restart, price change, PSS change, network change, printer change, or configuration change
- actions already taken and whether they changed the symptom
- exported diagnostics of all logs
- downloaded JOSN Support bundle

25. Field support boundaries

On a production DOMS, do not:

- request or use SSH as part of the normal support process
- open a terminal on the controller
- execute Node/npm scripts
- execute SQL manually
- edit package files
- create or edit `.env` files
- change operating-system service definitions
- disable TLS verification as a troubleshooting shortcut
- change prices as a connectivity test
- remap pumps/tanks by trial and error
- manipulate fiscal counters or receipt identifiers
- expose secrets in screenshots or support tickets

Use the VPOS configuration and diagnostic screens first. If the required correction is not exposed safely through the application or approved DOMS package-management workflow, escalate it as an engineering/deployment issue rather than bypassing the secure operating model.

26. Commissioning sign-off

Record the following in the deployment/change record before leaving site:

Item	Pass	Notes
Correct package for DOMS controller	☐	
Package installed through approved DOMS process	☐	
Application reachable and administrator login successful	☐	
Site profile verified	☐	
JPL settings tested	☐	
Forecourt Monitor stable	☐	
PSS/VPOS reconciliation accepted	☐	
Pumps/nozzles/tanks mapped correctly	☐	
Approved pricing verified	☐	
Printer tested	☐	
Proxy/fiscal path verified	☐	
Tanzania setup verified where applicable	☐	
Controlled transaction/receipt accepted	☐	
Controlled restart/persistence accepted	☐	
Manager training completed	☐	
Rollback package confirmed	☐	
Open issues documented	☐	

Technician: _____

Station: _____

DOMS controller: CPB-579 ☐ / CPB-539 ☐

Installed package/version: _____

Date/time: _____

Site representative: _____